CTI / v5.2.0.5

Clinical Trial Intelligence Workbench

Full operating & testing guide

Windows with Docker Desktop — PostgreSQL included

Source data, saved predictions and guided explanations

Research workflows · accuracy reports · acceptance checks

SET UP · IMPORT · PREDICT · REVIEW

Research decision support with source and model provenance

Editable Markdown accompanies this PDF. Use the PDF chapter bookmarks.

Historical approved models, raw records and metrics remain preserved.

Not clinical or regulatory advice. Final release evidence has its own report.

Docker operating manual edition · 19 October 2026

Contents

Begin with Windows Docker startup in chapter 2. Chapters 6–14 explain data and research use; chapters 15–22 cover testing, every section, files and evidence. Commands are provided in the accompanying Markdown for exact copying.

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Release v5.2.0.5. This guide covers setup, PostgreSQL, data, prediction, training, scientific explanations, reports, roles and testing. The supplied models run immediately; training is optional.

1. Start with the complete project

Download and extract the ZIP. Open the extracted folder containing `START_DOCKER_WINDOWS.bat`, `docker-compose.yml`, `backend`, `frontend`, `data` and `examples`. Do not run a launcher inside the ZIP viewer. Paths with spaces are supported.

The project includes a prebuilt interface, Python source, approved research artifacts, preserved registry responses, canonical Parquet shards, frozen experiments and this guide. PostgreSQL stores accounts, projects, permissions, saved predictions, conversations and storyboards. Private uploads remain under the application storage volume. SQLite is not used.

2. Windows with Docker Desktop

1. Open Docker Desktop and wait for its Linux-container engine to be running. Complete any WSL2/restart request.
2. Install Python 3.12 with the Windows launcher. Check `py -3.12 --version` in PowerShell.
3. Open PowerShell in the extracted project folder.
4. Run `.\START_DOCKER_WINDOWS.bat`.
5. Wait for image downloads, dependency installation, PostgreSQL health and application startup. The first build needs Internet access; it can take several minutes.
6. Open `http://localhost:8000` when the launcher reports the application ready.
7. Open the local `.env` in a text editor. Copy the value after `SETUP_TOKEN=` into the first-owner form. Enter your name, email and a unique password of at least 12 characters. Keep `.env` private.
8. Create a project. The genuine 101,135-record reference is selected for a new browser workspace. New projects use the new approved research models. Existing project selections are preserved.

```
py -3.12 --version
docker version
.\START_DOCKER_WINDOWS.bat
docker compose ps
```

Docker supplies PostgreSQL and the Python/Node build dependencies. A separate PostgreSQL installation and Node installation are unnecessary for this path. Keep Docker Desktop running while using the app.

3. Start, stop and upgrade safely

Use the same launcher for later starts. Stop the services with `docker compose down`; named volumes retain your work. Do not add `--volumes` or delete Docker volumes when you want to keep data.

```
docker compose down
docker compose up -d --build workbench
docker compose logs --tail 100 workbench postgres
```

Before upgrading an existing installation, back up PostgreSQL, the workbench-data volume and your private `.env`. Extract a new folder, review `.env.example`, and deliberately reuse the intended Compose project/volumes. A different folder name can create a separate empty Compose project. Do not overwrite an old installation blindly. Credential initialization preserves existing nonempty values; it does not reset accounts or database passwords.

4. Native Windows and Linux/macOS

The native path needs Python 3.12, a reachable PostgreSQL database and the supplied prebuilt UI. Node 22 is needed only to rebuild the frontend.

```
.\SETUP_WINDOWS.bat
.\CONNECT_DATABASE_WINDOWS.bat
.\START_WINDOWS.bat
```

Use the connection helper to set the native PostgreSQL connection, then start. For Linux/macOS, from the project root:

```
python3.12 scripts/setup.py --dev
.venv/bin/python scripts/configure_postgres.py
.venv/bin/python scripts/start.py
```

The single-process server serves the UI and API together. Use one worker: the verified public reference is cached once and occupies about 535 MB of deep DataFrame memory; model fitting needs additional memory. Containers should have sufficient RAM and disk for the source snapshot, models and build.

5. Environment values

`scripts/initialize_env.py` creates missing local PostgreSQL/setup credentials without printing them. `.env.example` contains names and nonsecret defaults. Never upload `.env`, session cookies, API keys or database credentials to GitHub or the assistant.

Setting	Purpose
`POSTGRES_USER`, `POSTGRES_DB`, `POSTGRES_PASSWORD`	Compose database account; generated password stays private
`DATABASE_URL`	Native PostgreSQL URL; Docker overrides the host for its service network
`COMPOSE_DATABASE_URL`	Database URL inside the Compose network
`TEST_DATABASE_URL`	Disposable test PostgreSQL database, separate from your research database
`SETUP_TOKEN`	First-owner bootstrap token; read it locally
`AUTH_MODE=session`	Account and project permissions
`MAX_TRAINING_ROWS=120000`	Bounded input-row ceiling; larger limits require deliberate resource planning
`MAX_DATASET_ROWS=100000`	Private-upload ceiling; the pinned public reference has its own verified count
`AI_PROVIDER`, provider API key and model ID	Optional assistant and scientific text drafts
`REQUIRE_EMAIL_VERIFICATION`, mail settings	Optional production email verification

Existing `.env` files may retain the older 20,000-row training limit. To train the large reference interactively, deliberately set `MAX_TRAINING_ROWS=120000`, then recreate the application container. This changes the input ceiling, not the number of eligible labels or available RAM.

After editing server environment values:

```
docker compose up -d --force-recreate workbench
```

6. Accounts, projects and roles

The first account owns its projects. Owners invite members under Access Control. Invitations carry a role; accepting an invitation does not grant owner permissions.

Role	Capabilities
Owner	Research, imports, scenarios, training, approval/activation, settings and membership
Trial researcher	Research, imports, scenarios and editable scientific explanations; cannot train or approve
Clinical reviewer	Read evidence and quality, review protocol evidence, use permitted assistant; scientific storyboards are read-only
Viewer	Read studies, predictions, approved quality and saved explanations; no writes or provider calls

Requests are project-scoped. Saved predictions, private models, uploads, conversations and storyboards are unavailable across unrelated projects. Refresh preserves the selected project and workspace route. Session expiry requires signing in again.

7. Understand the source data

The bundled official ClinicalTrials.gov API scope contains 101,135 unique completed interventional drug trials with reported Phase I–IV context, acquired on 18 October 2026. Preserved raw gzip responses and canonical shards have checksums. This is approximately one lakh acquired records, not one lakh fitted examples and not millions.

Historical exposure exclusions remove 2,610 source records. Duration eligibility requires exact ACTUAL observed start/completion dates and valid ordering. It leaves 41,968 records; outcome-availability purging removes 17,398 from the primary chronological experiment. Phase eligibility leaves 39,699 single-phase records with exact start-day precision. Partial dates and combined phases remain preserved in the source rather than being silently converted to exact dates or single labels.

Current registry snapshots can include post-start updates. Actual enrollment is not planned enrollment. The new design features populate planned enrollment only when the source explicitly says ESTIMATED. Source counts, fitted counts, tuning counts and test counts must be reported separately.

8. Import and inspect your own data

Open Data Manager to select a source, upload a supported table, import registry records, paste a table or use an allowed HTTPS source. Required columns are `nct_id` and `title`; useful fields include phase, enrollment, protocol text, design counts and reported dates/types. Review the validation/missingness report before interpreting results.

Imports do not train a model. A successful import opens Prediction Studio after the matching dataset overview is ready. The old-dataset request race has been corrected. The five sample records are software fixtures and cannot support scientific inference. The 800-record starter remains an optional historical compatibility cohort.

9. Overview and study exploration

Overview provides source counts, phase/status summaries, enrollment and graph choices. Its public-reference summary is derived from the complete source, with bounded display examples. Study Explorer supports title/ID search, phase/status filters, sorting and paging. Open a study to inspect its source, PCI and returned prediction. Compare up to three studies descriptively.

The initial study choices in analytical tools are bounded, not the entire cohort. Use Study Explorer to open another record. A cohort chart describes the selected source; model quality describes a model's labelled evaluation splits.

10. Predict without training

1. Open Prediction Studio.
2. Select Duration risk or Phase classification, the dataset and an approved model.
3. Run prediction for the selected/bounded rows.
4. Inspect returned values, input availability, model identity and warnings.
5. Reopen the saved batch to review its original model output. A later model activation does not rewrite that batch or substitute another task's result.

Retrospective mode accepts current source descriptors and labels its scope. Pretrial mode requires explicit availability timestamps for consumed features; supplied timestamps are not independent proof of prospective validity. Missing individual values use saved imputers; a missing required modality is rejected. Constant baselines consume no protocol features.

Duration is total start-to-completion time in days. Prolonged-duration probability refers to a threshold derived from the model's training durations. It is not trial success, adverse-event risk, or delay relative to an original planned schedule. Phase classification is consistency with a reported phase label, not prediction of phase progression.

11. Read actual model quality

Open Settings → Model quality, or Validation Lab. Choose the task and test/tuning/calibration split. This panel follows the active model for that task; use Models to change the active model with owner permission. Press Recheck saved evidence. The service verifies approved manifest/artifact hashes, validates saved-row membership and recomputes point metrics from the saved predictions. It does not fit or deserialize an estimator for this check.

Duration uses MAE, median absolute error, RMSE and R-squared. Risk uses AUROC, average precision, Brier/log loss and threshold classification support. Phase uses accuracy, balanced accuracy, macro-F1, per-class scores and confusion matrices. Charts are bounded deterministically; metrics use every validated row. Recorded bootstrap intervals are identified as recorded evidence, not newly recomputed intervals.

Unlabelled uploads do not have measurable accuracy. This panel never turns prediction confidence into model accuracy. A high majority-class percentage can conceal poor sensitivity.

12. Generate accuracy reports on your computer

With Docker running, double-click `GENERATE_MODEL_REPORT_DOCKER_WINDOWS.bat`. The helper exports the current release's bundled research model, copies HTML/JSON/CSV into the extracted folder's `reports` directory, and opens the HTML. Use the browser Print → Save as PDF for a local report PDF. The in-app Model quality download follows the project-selected model instead.

```
.\GENERATE_MODEL_REPORT_DOCKER_WINDOWS.bat --task duration
.\GENERATE_MODEL_REPORT_DOCKER_WINDOWS.bat --task phase
```

To export a specific historical bundle, add `--model-id bundled-duration-v5`. Native Python:

```
.\.venv\Scripts\python.exe scripts\generate_model_report.py --run-dir
  examples\registry-duration-v5205 --output-dir reports\duration-check
.\.venv\Scripts\python.exe scripts\generate_model_report.py --run-dir
  examples\registry-phase-refinement-v5205 --output-dir reports\phase-check
```

Use a new output directory. Reports preserve source files and label evaluation limitations.

13. Science Studio: content-based explanations

More tools → Science Studio provides a real interactive Three.js scene with captions and editable narration. Select a study using the study selector, wait for its evidence, then choose Use current research evidence. Dataset-only context uses a processing scene. Neuron, DNA or molecular scenes are chosen when the study topic supports them; a manual override is explicitly an educational metaphor.

The scene explains supplied content. It does not reconstruct patient anatomy, a measured molecule, genomic sequence, or the internal architecture of a tabular model. Actual source IDs, counts, dates, PCI and returned model values appear in source-derived caption steps. Missing values remain missing.

Edit title, source notes, steps, captions, narration and labels. Scene properties control accent/background, detail, camera, animation speed, labels and browser voice/rate. Play/pause, previous/next and reduced-motion controls remain available. Browser narration is opt-in and depends on installed voices; it stops on edits, backgrounding and navigation. Voice quality/offline availability varies by browser.

Save to the project, duplicate, load, delete, export/import bounded JSON, or export an actual rendered PNG with caption/labels. JSON transfer preserves AI-origin disclosures. Owners/researchers can edit; readers can explore/export saved explanations. Only declarative content is accepted—no imported scripts, remote assets or arbitrary generated code.

14. Optional AI drafts and assistant

Manual scientific editing and all core research features work without a provider key. Optional AI draft sends notes to a configured provider only after an explicit click. Configure an account-supported API model ID and server API key in `.env`; website subscriptions alone do not establish API access. Gemini/OpenAI/Claude/compatible routing is supported by the existing provider adapter.

AI-origin information, provider/model, original source hash and review status persist after edits and save/reload. Local draft receipts are project-scoped; imported claims without a matching receipt are marked declarations. Invalid/failed contacted-provider attempts count against the project request budget. A missing provider is detected before reserving that budget. No external provider success is claimed without a real configured call.

The research assistant provides readable streamed text, selected study/cohort/model context and source references. It distinguishes general explanations from project findings and does not invent missing records or live research access. Owner-reviewed feedback improves future prompt context; it does not silently retrain model weights. Project/operator/provider budgets still apply; unlimited correct answers cannot be guaranteed.

15. Analytical animations, scenarios and timelines

AI Pipeline and Data Fusion explain recorded inputs, transformations and outputs. Explainability labels local attribution, global importance, constant predictors and unavailable SHAP separately; SHAP is not causality. Recruitment shows source-supported stages and leaves absent patient-flow counts unavailable.

Trial Timeline preserves date precision/types, distinguishes observed duration from predicted total duration and avoids inventing treatment milestones or planned dates. Phase Academy explains Phase I–IV objectives conceptually; its illustrations are not patient observations.

What-If Lab and Scenarios allow supported protocol changes and rerun a source-compatible model. Operational planning assumptions are separately labelled. They do not become validated recruitment probabilities. Missing stage timing remains missing. The 2D evidence remains readable if WebGL is unavailable or reduced motion is selected.

16. Train and compare a project candidate

Owners open Experiments or Phase Model, review readiness and select candidate options. Standard/expanded CatBoost and new design candidates are optional. MAE or RMSE selects the duration objective on tuning data; risk and phase have explicit separate selection rules. One server training job runs at a time.

Readiness uses the requested configuration and the same eligibility/split guards as launch. It shows actual train/tuning/calibration/test counts and a concrete reason when blocked. If the large source exceeds an older operator limit, update that limit deliberately; do not duplicate or drop rows just to pass a check.

Training creates a candidate, metrics, row predictions, exclusions and manifests. It never replaces an existing active model automatically. Review test scores and limitations in Models; owners approve a research candidate and explicitly activate it per task. Keep the final test separate from hyperparameter selection. Repeatedly tuning to a previously inspected test invalidates an untouched-test claim.

17. Reproduce the larger frozen experiment

The release preserves raw acquisition and frozen prepared data. Verification commands never fit:

```
.\.venv\Scripts\python.exe scripts\acquire_registry_cohort.py --output data\registry-v5205
  --verify-only
.\.venv\Scripts\python.exe scripts\build_registry_runtime_manifest.py --check
.\.venv\Scripts\python.exe scripts\train_reference_models.py --prepared
  data\registry-v5205-training --output examples\registry-v5205
.\.venv\Scripts\python.exe scripts\verify_registry_models.py --output
  reports\registry-replay.json
.\.venv\Scripts\python.exe scripts\verify_v5_models.py --output reports\historical-replay.json
```

The training runner defaults to prefit checks. `--execute` explicitly fits. Use a new output directory for a separate experiment; the preserved original output is not overwritten. `--resume` skips hash-verified completed tasks and refuses an incomplete attempted fit. Source/config/protocol/code hashes and exact partition IDs are frozen before fitting. `scripts/export_reference_metrics.py` creates a full scalar CSV from unchanged original metrics; it does not fit or reevaluate models.

Acquiring a different scope requires a deliberately new dataset/output/protocol. The current source query has about one lakh records; it cannot supply millions by duplication. Millions of rows would also require a different compute and validation design.

18. Primary measured evidence

The larger frozen duration experiment uses 8,806 train, 4,624 tuning, 2,841 calibration and 8,299 test trials. The threshold is 1,105 days. Phase uses 16,932 train, 14,634 tuning and 8,133 test trials. Historical IDs are disjoint. Current snapshots and completed-only selection still limit interpretation.

Task/metric	Matched baseline	Tuning-selected result
Duration test MAE, days	285.5099	244.2604
Duration test RMSE, days	331.7757	313.7106
Duration test R-squared	-0.1075	0.0098
Risk test AUROC	0.5000	0.8151
Risk test average precision	0.0321	0.1159
Risk test Brier, lower better	0.07845	0.05257
Phase test accuracy	22.4271%	73.7489%
Phase test balanced accuracy	25.0000%	68.0089%
Phase test macro-F1	0.09159	0.67683

At the fixed 0.5 risk threshold, accuracy is 94.3728%, but sensitivity is 20.6767%, precision 17.6849% and balanced accuracy 58.7449%. The always-negative baseline is 96.7948% accurate because only 266/8,299 test records are prolonged. Therefore “over 90% accuracy” alone would be misleading. Raw probabilities are primary; calibration is unavailable here because the reserved calibration split has no positives. Empirical conformal coverage is not a guarantee under temporal shift.

The duration serving pair uses the frozen independently selected duration/risk components, not the candidate with the lowest observed test error. New projects receive these manually reviewed research entries; older project activations remain unchanged. The separately evaluated phase refinement below is the default for new projects. The original exact-date phase model remains selectable, and neither evaluation is relabelled as the other.

Separately evaluated phase refinement

After the original test was inspected, three fixed candidates were fitted on the same 16,932 training records and selected using only the original 14,634 tuning records. Word/character TF-IDF plus logistic regression won tuning macro-F1. No original test records entered this fit or candidate selection.

A frozen, ID-disjoint group of 50,139 pure-phase records with partial or unavailable exact start dates was then audited once. It excludes every original phase role and historical-exposure ID.

Both the old and refined classifier were measured on this same group:

Matched subgroup metric	Original classifier	Refined classifier
Accuracy	66.4752%	67.9312%
Balanced accuracy	65.2497%	67.3369%
Macro-F1	0.65740	0.67400

The paired accuracy difference is +1.4560 percentage points, with a conditional trial-bootstrap 95% interval of +1.1657 to +1.8590 points. This subgroup differs in date-reporting characteristics; it is not a chronological or prospective test. Do not compare its 67.93% directly with 73.75% on the original, different 8,133-record test. There are no supported EARLY_PHASE1 examples in this audit, so no class-specific accuracy claim is made for that label.

New projects default to `registry-phase-refinement-v5205`. Existing projects retain their activation choices. The original phase model remains available. The two additional risk candidates did not beat the selected original risk model on tuning Brier score; they are not activated, and their proposed fresh stress audit was skipped before new labels were accessed. There is no verified 90% phase-accuracy result.

`examples/refinement-v5205-evidence` preserves frozen protocols/IDs, development candidates, tuning predictions and audit reports. The portable wrapper below defaults to a dry run. Add `--execute` only for a deliberate fit/replay, and use a new output directory. Replaying this already inspected audit is reproducibility, not a new independent validation.

```
.\.venv\Scripts\python.exe scripts\run_refinement.py --task phase --output results\phase-refit
.\.venv\Scripts\python.exe scripts\run_refinement.py --task risk --output results\risk-refit
.\.venv\Scripts\python.exe scripts\run_refinement.py --task phase-audit --output
  results\phase-audit-replay
```

The application honors the approved refinement manifest's empty narrative-slot contract before text truncation. A custom Python integration must apply `backend.app.services.phase_service.normalize_phase_input` before calling the low-level phase predictor for this artifact.

19. Every application section

Section	Start and check
Overview	Select source, change a chart, compare displayed cohort counts
Studies	Search/filter/page, open a source record
Prediction Studio	Predict with an approved saved artifact; reopen original batch
Protocol Evidence	Upload/review supported document evidence in your project
Scenarios	Change supported assumptions and inspect labelled model/planning results
Models	Review provenance/metrics/gates; owner approval and activation
Assistant	Configure provider, ask a source-grounded question, check references
Settings	Appearance/motion, project controls and real Model quality
Science Studio	Load actual evidence, edit/explore/save/export a scientific explanation
Research home	Enter a research workflow and inspect conceptual scientific introduction
Research sources / Saved library	Find/save source records with actual provenance
Research Hub	Inspect selected study context and retrieval limits
Complexity Lab	Review descriptive PCI and missing-component handling
Formula Studio	Inspect explicit formulas and supported parameters
Experiments	Owner readiness, fitting, metrics, reports and preserved history
Validation Lab	Saved-model quality, diagnostics and completed project experiments
Phase Academy	Conceptual phase objectives and evidence distinctions
Phase Model	Consistency classification and owner training
Data Manager	Select/import/validate source records
AI Pipeline	Step through supported processing and outputs
Trial Timeline	Check source dates/precision and predicted total-duration context
Explainability	Inspect the actual explanation method; unavailable SHAP remains unavailable
Recruitment	Inspect only reported stage counts and assumptions
What-If Lab	Rerun supported model inputs; compare actual returned outputs
Data Fusion	Inspect consumed modalities and composite branches
Access Control	Invite members and verify role restrictions

20. Automated testing

Tests require a disposable PostgreSQL database. Never use your research database for `TEST_DATABASE_URL`. Native test setup installs development dependencies; Docker's application image contains runtime dependencies, not the full development test environment.

```
py -3.12 scripts\setup.py --dev
docker compose --profile test up -d postgres-test
.\TEST_WINDOWS.bat
```

The test helper reads `TEST_DATABASE_URL` from your private `.env` and uses disposable private schemas. The Compose test database is published on port 5433 by default; wait for its health check before testing. You can pass normal pytest selectors to `TEST_WINDOWS.bat`. Frontend development checks:

```
cd frontend
npm ci
npm run build
cd ..
node scripts\check_stream_parser.mjs
```

The release verification report records exactly which checks ran, skips, browser evidence and platform limits. A syntax check alone is not functional verification. Windows GUI execution and audible browser voice availability are not inferred from Linux tests.

21. Manual acceptance test

Create/sign in → create project → confirm genuine dataset → Overview count 101135 → open a study → predict both tasks without fitting → reopen saved prediction → change model and confirm old batch unchanged → Settings Model quality → recheck/download both reports → Science Studio load evidence → edit/animate/save/reload/PNG/JSON → refresh and confirm same route/project → try reviewer/viewer restrictions → run a small eligible owner experiment only if desired → review before approval/activation → stop/restart and confirm persistence.

Also test 390-pixel mobile navigation, keyboard controls, reduced motion, missing WebGL and provider-disabled states. Unsupported data, missing model inputs and stale storyboard saves should produce actionable errors, not invented values.

22. Troubleshooting

If Docker reports `lookup auth.docker.io: no such host`, it cannot resolve Docker Hub. This occurs before project code runs. Check the computer's connection, VPN/proxy policy and Docker Desktop network/proxy settings. These PowerShell checks help locate the issue:

```
Resolve-DnsName auth.docker.io
Test-NetConnection auth.docker.io -Port 443
docker pull python:3.12-slim
docker pull node:22-bookworm-slim
```

Restart Docker Desktop after correcting its network configuration, then rerun the launcher. Keep TLS/certificate verification enabled. Offline first builds cannot download uncached base images.

If port5432 is occupied, set `POSTGRES_PORT=5434` in `.env` before starting. If port8000 is occupied, stop the other application or deliberately change the Compose publication. For database login errors, verify that the password matches the existing database volume; changing an environment variable does not reset an initialized PostgreSQL account.

For UI/API connection problems, check `docker compose ps` and the service logs. For training, read the readiness reason and configured row/memory limits. For a missing AI reply, verify API access, model ID, quota and provider status; core features remain available. For 3D problems, choose 2D or low detail; captions/evidence remain readable. Restore a modified model/reference file from the verified release rather than disabling hash checks.

23. Files, backup and research limits

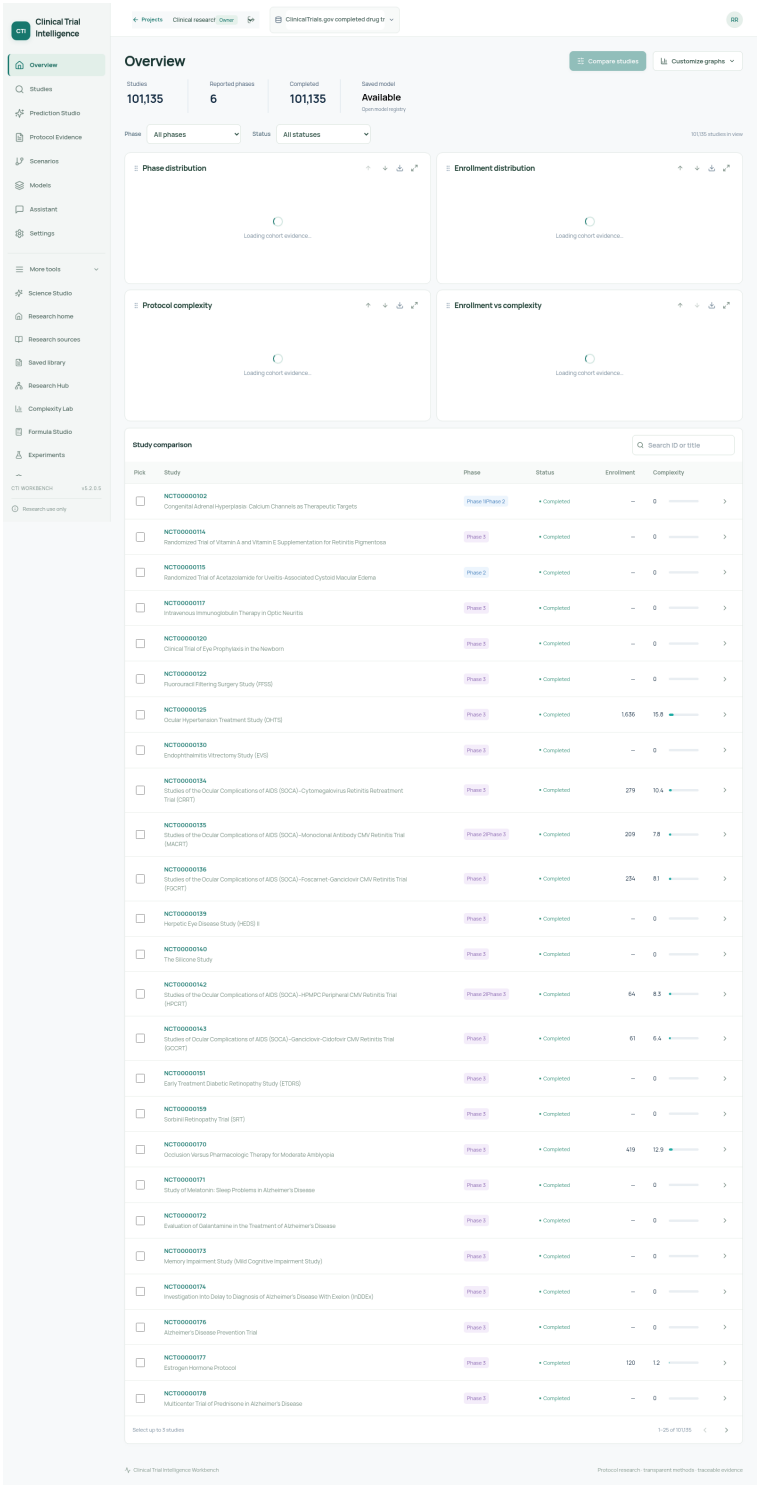
`backend/app/platform.py` enforces accounts/project access; `v5_inference.py` resolves approved models and saved batches. `science_studio.py` stores bounded project storyboards. `model_quality_routes.py` exposes verified saved-row analysis. `src/clinical_trial_intelligence` contains training/inference/research methods. `data/registry-v5205` preserves the acquired scope; `data/registry-v5205-training` records exclusions and frozen partitions; `examples/registry-v5205` preserves the original fitted runs. `examples/registry-duration-v5205` and `registry-phase-v5205` are exact serving copies. Historical data/artifacts remain preserved.

Back up PostgreSQL, workbench-data/results volumes, private uploads and `.env` securely. Source ZIPs exclude runtime secrets and your private project data. Exported source/model reports are research artifacts, not clinical or regulatory validation. No perfect accuracy, automatic learning from chat, validated patient outcome model, or million-record training run is claimed.

24. Actual application screenshots

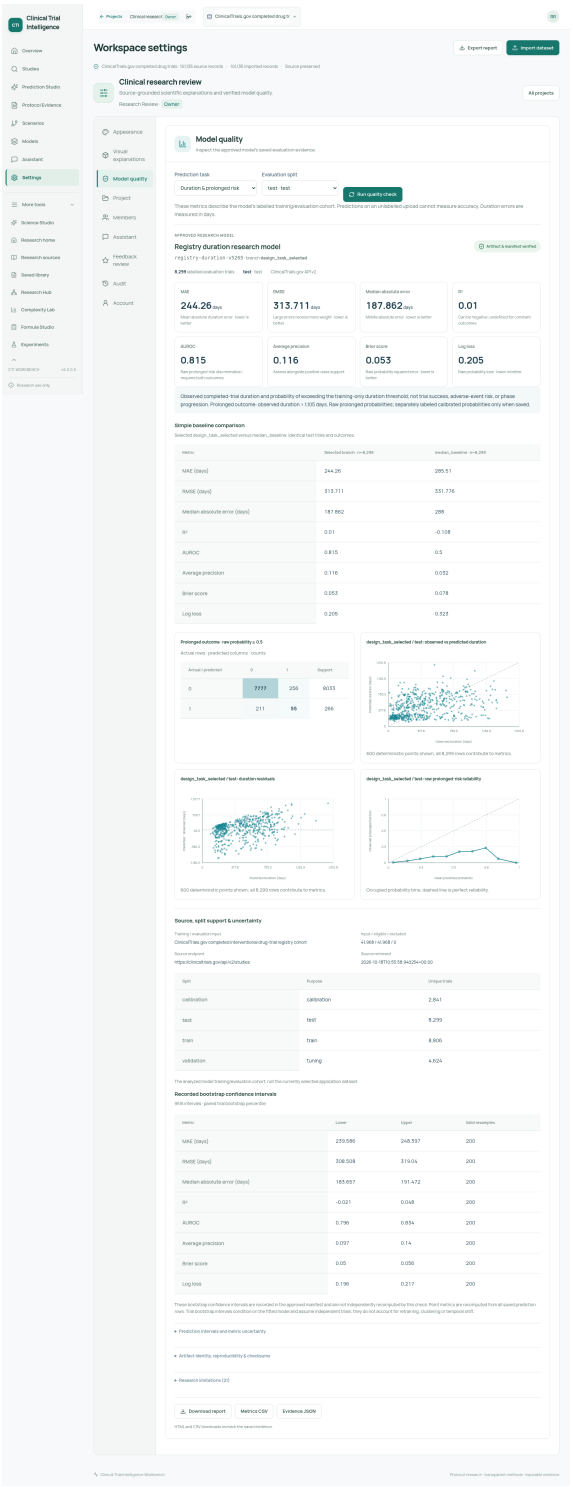
Screenshots captured from the application during release verification. Read the accompanying report for exercised workflows and limits.

Overview · full-cohort research workspace



overview-desktop.png

Duration model quality · saved predictions and recomputed metrics



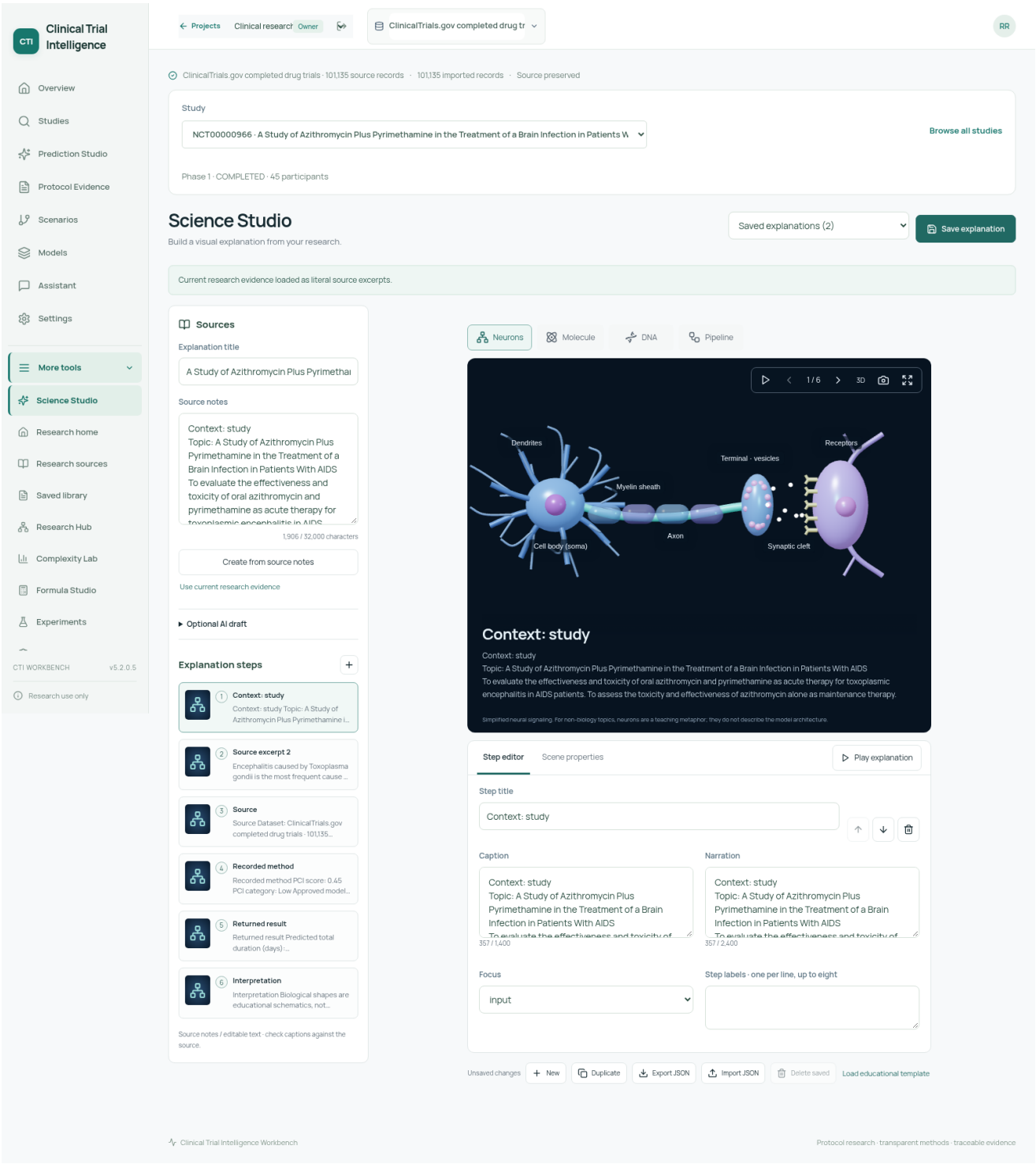
model-quality-duration.png

Refined phase quality · matched subgroup evidence



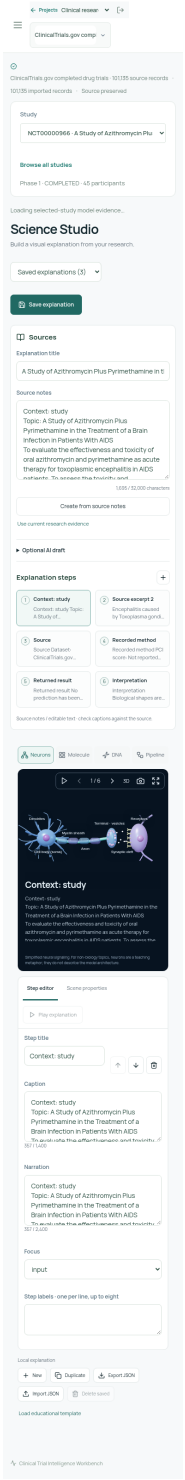
model-quality-phase-refinement.png

Science Studio · actual selected-study content and editable neurons



science-project.png

Mobile · responsive scientific explanation



science-project-mobile.png